ELECTRODYNAMICS II · SEMESTER 5

Induction and Maxwell equations

Module 2 · Answer key

40 worked answers. Each answer names the prepare topic to revise first. Mark splits are suggested study rubrics.

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Section A

Q1 Model · syllabus-based

State local Ohm's law, define conductivity and distinguish it from resistivity.

KNOW FIRST

T1 Ohm's law in a material P1 School physics bridge M1 Mathematics from the beginning

For an isotropic conductor at rest, $\mathbf{J} = \sigma \mathbf{E}$. Conductivity σ has units S/m; resistivity is $\varrho = 1/\sigma$ in Ω m. It is an empirical linear response, not a universal law for all materials.

ANSWER $J = \sigma E$; $\varrho = 1/\sigma$.

Marks: 1 mark: correct physical statement; 1 mark: explanation or calculation; 1 mark: required result with condition/unit.

Q2 Model · syllabus-based

Derive the resistance of a uniform wire of length ℓ and area A from local Ohm's law.

KNOW FIRST

T1 Ohm's law in a material P1 School physics bridge M1 Mathematics from the beginning

Use $J = I/A$ and $E = V/\ell$. Then $I/A = \sigma V/\ell$, giving $V/I = \ell/(\sigma A)$. The assumptions are uniform conductivity, cross-section and steady current.

ANSWER $R = \ell/(\sigma A)$.

Marks: 1 mark: correct physical statement; 1 mark: explanation or calculation; 1 mark: required result with condition/unit.

Q3 Model · syllabus-based

Define emf and explain why its name can be misleading.

KNOW FIRST

T2 Electromotive force P1 School physics bridge
M1 Mathematics from the beginning

Emf is closed-path driving work per unit charge, $\mathcal{E} = \oint \mathbf{f} \cdot d\mathbf{l}$. It has units J/C = volt. It is not a mechanical force and need not equal the loaded source's terminal voltage.

ANSWER **Emf measures energy per charge, in volts.**

Marks: 1 mark: correct physical statement; 1 mark: explanation or calculation; 1 mark: required result with condition/unit.

Q4 Model · syllabus-based

Can a purely electrostatic field provide a nonzero emf around a closed loop? Explain.

KNOW FIRST

T2 Electromotive force P1 School physics bridge
M1 Mathematics from the beginning

No. $\mathbf{E} = -\nabla V$ for a single-valued electrostatic potential, so $\oint E \cdot dl = 0$. A battery needs a non-electrostatic source force; an induced electric field is not electrostatic.

ANSWER **Purely electrostatic closed-loop emf is zero.**

Marks: 1 mark: correct physical statement; 1 mark: explanation or calculation; 1 mark: required result with condition/unit.

Q5 Model · syllabus-based

A 0.40 m rod moves at 3.0 m/s perpendicular to a 0.50 T field and to its own length. Find the emf magnitude.

KNOW FIRST

T3 Motional emf and energy supply P1 School physics bridge M1 Mathematics from the beginning

$\mathcal{E} = B\ell v = (0.50)(0.40)(3.0) = 0.60$ V. Charge separation is along $\mathbf{v} \times \mathbf{B}$; the positive end depends on the stated vector directions.

ANSWER $\mathcal{E} = 0.60$ **V.**

Marks: 1 mark: correct physical statement; 1 mark: explanation or calculation; 1 mark: required result with condition/unit.

Q6 Model · syllabus-based

If magnetic forces do no work on individual charges, where does the heat in a moving-rod circuit come from?

KNOW FIRST

T3 Motional emf and energy supply P1 School physics bridge M1 Mathematics from the beginning

The pulling agent works against magnetic drag on the current-carrying rod. In steady motion its mechanical power equals the resistor's heating

power. Magnetic forces redirect motion but supply no net work on a charge.

ANSWER **The external mechanical agent supplies the energy.**

Marks: 1 mark: correct physical statement; 1 mark: explanation or calculation; 1 mark: required result with condition/unit.

Q7 Model · syllabus-based

State Lenz's rule and distinguish opposition to flux from opposition to flux change.

KNOW FIRST

T4 Faraday's law and Lenz's rule P1 School physics bridge M1 Mathematics from the beginning

The induced current produces magnetic effects opposing the change in linked flux. If an existing flux decreases, the induced field acts in the same direction as that flux to oppose its decrease.

ANSWER **The opposition is to change, not always to the field itself.**

Marks: 1 mark: correct physical statement; 1 mark: explanation or calculation; 1 mark: required result with condition/unit.

Q8 Model · syllabus-based

List three ways to change the flux through a flat loop in uniform B.

KNOW FIRST

T4 Faraday's law and Lenz's rule P1 School physics bridge M1 Mathematics from the beginning

Since $\Phi_B = BA\cos\theta$, changing B, loop area A or orientation θ can change flux. θ is measured from the area normal. A nonzero flux does not by itself imply an emf.

ANSWER **Emf requires a time derivative of linked flux.**

Marks: 1 mark: correct physical statement; 1 mark: explanation or calculation; 1 mark: required result with condition/unit.

Q9 Model · syllabus-based

Can there be induced E at a location containing no conductor? Give the governing law.

KNOW FIRST

T5 Induced electric field P1 School physics bridge M1 Mathematics from the beginning

Yes. $\nabla \times E = -\partial B/\partial t$ describes a field in space. A wire merely provides mobile charges that respond; the induced E exists without it.

ANSWER **Changing magnetic flux produces electric circulation without a wire.**

Marks: 1 mark: correct physical statement; 1 mark: explanation or calculation; 1 mark: required result with condition/unit.

Q10 Model · syllabus-based

Why cannot a general induced E be described using only a single-valued scalar electrostatic potential?

KNOW FIRST

T5 Induced electric field P1 School physics bridge M1 Mathematics from the beginning

A gradient has zero curl and zero closed-loop integral. A changing B gives $\oint E \cdot dl = -d\Phi_B/dt$, which can be nonzero. Thus $E = -\nabla V$ alone is insufficient.

ANSWER Nonzero circulation rules out a purely scalar electrostatic description.

Marks: 1 mark: correct physical statement; 1 mark: explanation or calculation; 1 mark: required result with condition/unit.

Q11 Model · syllabus-based

Define self-inductance and the henry, stating the linearity assumption.

KNOW FIRST

T6 Self and mutual inductance P1 School physics bridge M1 Mathematics from the beginning

For a fixed linear circuit, flux linkage is $N\Phi = LI$. One henry corresponds to one weber-turn per ampere, or an induced emf magnitude of one volt when current changes at one ampere per second.

ANSWER $L = N\Phi/I$ in henries for a linear circuit.

Marks: 1 mark: correct physical statement; 1 mark: explanation or calculation; 1 mark: required result with condition/unit.

Q12 Model · syllabus-based

Distinguish the induced emf of an ideal inductor from its passive terminal voltage drop.

KNOW FIRST

T6 Self and mutual inductance P1 School physics bridge M1 Mathematics from the beginning

The induced emf opposes the current change: $\mathcal{E}_L = -L\,dI/dt$. With current entering the positive-labelled terminal, the passive drop is $v_L = +L\,dI/dt$. They describe the same induction with different sign conventions.

ANSWER **Back emf is negative relative to increasing current; passive drop is positive.**

Marks: 1 mark: correct physical statement; 1 mark: explanation or calculation; 1 mark: required result with condition/unit.

Q13 Model · syllabus-based

Define mutual inductance and explain why coil orientation matters.

KNOW FIRST

T6 Self and mutual inductance P1 School physics bridge M1 Mathematics from the beginning

Linkage in coil 2 due to I_1 is $\lambda_{21} = MI_1$. Then $\mathcal{E}_2 = -M\,dI_1/dt$. Reversing one coil's reference normal changes the signed linkage and M, while its coupling magnitude is unchanged.

ANSWER **State winding directions or use a consistent dot convention.**

Marks: 1 mark: correct physical statement; 1 mark: explanation or calculation; 1 mark: required result with condition/unit.

Q14 Model · syllabus-based

Find the energy of a 0.30 H inductor carrying 2.0 A. What happens if current is reversed without changing magnitude?

KNOW FIRST

T7 Magnetic field energy P1 School physics bridge M1 Mathematics from the beginning

$U = LI^2/2 = (0.30)(4)/2 = 0.60$ J. Since energy depends on I^2, reversing I changes field direction but not stored energy for an ideal linear inductor.

ANSWER $U = 0.60$ **J for either current direction.**

Marks: 1 mark: correct physical statement; 1 mark: explanation or calculation; 1 mark: required result with condition/unit.

Q15 Model · syllabus-based

Write and interpret the local continuity equation.

KNOW FIRST

T8 Charge conservation and Maxwell's repair P1 School physics bridge M1 Mathematics from the beginning

$\partial\rho/\partial t + \nabla \cdot J = 0$. A positive net outward current from a tiny volume reduces its charge density. This is local conservation of electric charge.

ANSWER **Charge accumulation balances net inward current.**

Marks: 1 mark: correct physical statement; 1 mark: explanation or calculation; 1 mark: required result with condition/unit.

Q16 Model · syllabus-based

What is displacement current in vacuum? Does it require charge carriers crossing the surface?

KNOW FIRST

T8 Charge conservation and Maxwell's repair P1 School physics bridge M1 Mathematics from the beginning

Its density is $J_d = \epsilon_0 \partial E/\partial t$, and its integrated contribution is $I_d = \epsilon_0 d\Phi_E/dt$. It can occur through a vacuum capacitor gap with no conduction carriers crossing.

ANSWER **Displacement current is a changing-field contribution to magnetic circulation.**

Marks: 1 mark: correct physical statement; 1 mark: explanation or calculation; 1 mark: required result with condition/unit.

Q17 Prerequisite-course PYQ A26 Q15 · adapted marks

Show that the divergence of a magnetic field vanishes, and state its integral form.

KNOW FIRST

T9 The four Maxwell equations P1 School physics bridge M1 Mathematics from the beginning

$\oint B \cdot da = 0$ and $\nabla \cdot B = 0$. Writing $B = \nabla \times A$ gives $\nabla \cdot B = \nabla \cdot (\nabla \times A) = 0$ because mixed derivatives cancel for a smooth vector potential. Thus magnetic field lines have no net source or sink: as much flux enters a closed surface as leaves it.

ANSWER **No magnetic monopole source appears in these equations.**

Marks: 1 mark: correct physical statement; 1 mark: explanation or calculation; 1 mark: required result with condition/unit.

Q18 Model · syllabus-based

Write Faraday's law in differential form and explain each side.

KNOW FIRST

T9 The four Maxwell equations P1 School physics bridge M1 Mathematics from the beginning

$\nabla \times E = -\partial B/\partial t$. The left side measures local electric circulation per area; the right side gives the opposing rate of magnetic-field change.

ANSWER **Changing B generates circulating E.**

Marks: 1 mark: correct physical statement; 1 mark: explanation or calculation; 1 mark: required result with condition/unit.

Q19 Model · syllabus-based

State Ampère-Maxwell law in matter and identify the sources on its right side.

KNOW FIRST

T10 Maxwell equations in matter P1 School physics bridge M1 Mathematics from the beginning

$\nabla \times H = J_f + \partial D/\partial t$. The first term is free conduction/current density; the second includes vacuum and polarization response because $D = \epsilon_0 E + P$.

ANSWER **The H curl is driven by free current plus changing D.**

Marks: 1 mark: correct physical statement; 1 mark: explanation or calculation; 1 mark: required result with condition/unit.

Q20 Model · syllabus-based

Why is normal B continuous at an ordinary interface, even if permeability changes?

KNOW FIRST

T11 Electromagnetic boundary conditions P1 School physics bridge M1 Mathematics from the beginning

A thin pillbox and $\oint B \cdot da = 0$ give $B_{2n} = B_{1n}$. Permeability affects the corresponding H components, but there is no magnetic surface charge in Maxwell's equations.

ANSWER **Normal B is continuous.**

Marks: 1 mark: correct physical statement; 1 mark: explanation or calculation; 1 mark: required result with condition/unit.

Section B

Q21 Model · syllabus-based

A wire has ℓ=2 m, A=1.0×10^{-6} m² and σ=5.0×10^{7} S/m. Apply 0.10 V. Find R, I, E, J and total power.

KNOW FIRST

T1 Ohm's law in a material P1 School physics bridge M1 Mathematics from the beginning

1. $R = \ell/(\sigma A) = 0.040\ \Omega$.
   Why: The wire is uniform and ohmic.
2. $I = V/R = 2.5$ A and $E = V/\ell = 0.050$ V/m.
   Why: Use the terminal voltage along its length.
3. $J = \sigma E = 2.5 \times 10^{6}$ A/m², also I/A.
   Why: Local and global current descriptions must agree.
4. $P = VI = 0.25$ W; equivalently $JE(A\ell) = 0.25$ W.
   Why: Field power density integrates to the circuit power.

ANSWER $R = 0.040\ \Omega$, I=2.5 A, E=0.050 V/m, $J = 2.5 \times 10^{6}$ A/m², P=0.25 W.

Marks: 1 mark: law and assumptions; 4 marks: explanation or calculation; 1 mark: conclusion with units/limiting check.

Q22 Model · syllabus-based

A 12 V source of internal resistance 1 Ω drives a 5 Ω load. Find current, terminal voltage and the power budget.

KNOW FIRST

T2 Electromotive force P1 School physics bridge M1 Mathematics from the beginning

1. The closed-loop equation is $12 = I(1 + 5)$.
   Why: Both resistances carry the same current.
2. I=2 A and $V_{\text{term}} = 12 - Ir = 10$ V.
   Why: Internal voltage drop reduces the external voltage.
3. Source power is $\mathcal{E}I = 24$ W; internal heating is $I^2 r = 4$ W and load power is $I^2 R = 20$ W.
   Why: Energy supplied equals total dissipation.

ANSWER **I=2 A, terminal voltage 10 V; 24 W supplied = 4 W internal + 20 W load.**

Marks: 1 mark: law and assumptions; 4 marks: explanation or calculation; 1 mark: conclusion with units/limiting check.

Q23 Model · syllabus-based

The rod in Q5 completes a circuit of total resistance 2 Ω. Find current, magnetic drag and pulling power at constant speed.

KNOW FIRST

T3 Motional emf and energy supply P1 School physics bridge M1 Mathematics from the beginning

1. $I = \mathcal{E}/R = 0.60/2 = 0.30$ A.
   Why: The circuit is resistive and induction transients are neglected.

2. $F = BI\ell = (0.50)(0.30)(0.40) = 0.060$ N opposite motion.
   Why: Lenz's law fixes the drag direction.

3. $P_{\text{pull}} = Fv = 0.18$ W and $I^2R = 0.18$ W.
   Why: Steady motion leaves no kinetic-energy increase.

ANSWER **I=0.30 A, drag=0.060 N, input power=0.18 W equals heating.**

Marks: 1 mark: law and assumptions; 4 marks: explanation or calculation; 1 mark: conclusion with units/limiting check.

Q24 Model · syllabus-based

A 200-turn coil of area 0.010 m² has its normal along a field that rises uniformly from 0.10 T to 0.40 T in 0.050 s. Find its emf, including sign.

KNOW FIRST

T4 Faraday's law and Lenz's rule P1 School physics bridge M1 Mathematics from the beginning

1. Choose positive normal along B, so $\Phi_B = BA$.
   Why: The angle is zero.

2. $dB/dt = 0.30/0.050 = 6.0$ T/s.
   Why: The ramp is uniform.

3. $\mathcal{E} = -NA\,dB/dt = -200(0.010)(6.0) = -12$ V.
   Why: Use total linkage rather than one-turn flux.

ANSWER **−12 V in the positive-loop orientation; magnitude 12 V.**

Marks: 1 mark: law and assumptions; 4 marks: explanation or calculation; 1 mark: conclusion with units/limiting check.

Q25 Model · syllabus-based

A coil has N turns and area A and rotates with angular speed ω in uniform B. Derive its sinusoidal emf when its normal initially points along B.

KNOW FIRST

T4 Faraday's law and Lenz's rule P1 School physics bridge M1 Mathematics from the beginning

1. The angle is $\theta = \omega t$, giving linkage $\lambda = NAB\cos\omega t$.
   Why: Rotation changes orientation, not B or A.
2. Differentiate: $d\lambda/dt = -NAB\omega\sin\omega t$.
   Why: The chain rule contributes ω.
3. Faraday's minus sign gives $\mathcal{E} = NAB\omega\sin\omega t$.
   Why: The emf opposes the signed flux change.
4. At maximum linkage the emf is zero; at zero linkage its magnitude is maximum.
   Why: A slope is not the same as the function value.

ANSWER **Peak emf $\mathcal{E}_0 = NAB\omega$.**

Marks: 1 mark: law and assumptions; 4 marks: explanation or calculation; 1 mark: conclusion with units/limiting check.

Q26 Model · syllabus-based

Uniform $B(t)\hat{z}$ occupies a circular region of radius a. Derive the azimuthal E for r<a, with positive circulation counterclockwise viewed from +z.

KNOW FIRST

T5 Induced electric field P1 School physics bridge M1 Mathematics from the beginning

1. Choose a concentric circle of radius r.
   Why: Axial symmetry makes E tangential and constant.
2. The enclosed flux is $B\pi r^2$.
   Why: The whole surface lies within the field region.
3. $E_\phi(2\pi r) = -\pi r^2 \dot{B}$.
   Why: Apply Faraday's law with the right-hand normal +z.
4. Thus $E_\phi = -(r/2)\dot{B}$.
   Why: Divide by circumference.

ANSWER Inside: $\mathbf{E} = -(r/2)\dot{B}\,\hat{\boldsymbol{\phi}}$.

Marks: 1 mark: law and assumptions; 4 marks: explanation or calculation; 1 mark: conclusion with units/limiting check.

Q27 Model · syllabus-based

For Q26 find E at r>a and evaluate its magnitude for a=0.10 m, r=0.20 m and $\dot{B} = 4$ T/s.

KNOW FIRST

T5 Induced electric field P1 School physics bridge M1 Mathematics from the beginning

1. Only the disk contributes flux, so $\Phi = B\pi a^2$.
   Why: B is zero outside the disk.
2. $2\pi r E_\phi = -\pi a^2 \dot{B}$, hence $E_\phi = -a^2\dot{B}/(2r)$.
   Why: The path circumference still uses r.
3. The magnitude is $(0.10)^2(4)/(2 \times 0.20) = 0.10$ V/m.
   Why: Substitute in SI units.
4. The inside and outside formulas agree at r=a.
   Why: This checks continuity at the boundary.

ANSWER Outside: $E_\phi = -a^2\dot{B}/(2r)$; magnitude 0.10 V/m here.

Marks: 1 mark: law and assumptions; 4 marks: explanation or calculation; 1 mark: conclusion with units/limiting check.

Q28 Model · syllabus-based

Derive L for a long solenoid with N turns, length ℓ, area A and uniform linear permeability μ.

KNOW FIRST

T6 Self and mutual inductance P1 School physics bridge M1 Mathematics from the beginning

1. Inside, $H = NI/\ell$ and $B = \mu NI/\ell$.
   Why: Neglect end effects and use the free-current circulation law.

2. Each turn links $\Phi = BA$.
   Why: The field is approximately uniform over the cross-section.

3. Total linkage is $N\Phi = \mu N^2 AI/\ell$.
   Why: N identical turns link the same flux.

4. Divide linkage by current.
   Why: Inductance is defined by $\lambda = LI$.

ANSWER $L = \mu N^2 A/\ell$.

Marks: 1 mark: law and assumptions; 4 marks: explanation or calculation; 1 mark: conclusion with units/limiting check.

Q29 Model · syllabus-based

Two ideal coaxial long coils share area 2×10^{-4} m² and length 0.20 m, with 500 and 1000 turns. In air, find M and the emf magnitude in coil 2 when $dI_1/dt = 3$ A/s.

KNOW FIRST

T6 Self and mutual inductance P1 School physics bridge M1 Mathematics from the beginning

1. Coil 1 makes $B_1 = \mu_0 N_1 I_1/\ell$.
   Why: Assume all of its uniform interior flux links coil 2.

2. $\lambda_{21} = N_2 B_1 A = MI_1$, so $M = \mu_0 N_1 N_2 A/\ell$.
   Why: Use the common cross-sectional area.

3. $M = 6.283 \times 10^{-4}$ H.
   Why: Substitute the specified geometry.

4. $|\mathcal{E}_2| = M|\dot{I}_1| = 1.885 \times 10^{-3}$ V.
   Why: Faraday's law differentiates linkage.

ANSWER **M=0.6283 mH; induced emf magnitude 1.885 mV.**

Marks: 1 mark: law and assumptions; 4 marks: explanation or calculation; 1 mark: conclusion with units/limiting check.

Q30 Model · syllabus-based

Derive the stored energy of a constant-L inductor from the work supplied by a source.

KNOW FIRST

T7 Magnetic field energy P1 School physics bridge M1 Mathematics from the beginning

1. The source balances passive inductor drop $v = L\,dI/dt$.
   Why: It must overcome the opposing induced emf.

2. Power is $p = vI = LI\,dI/dt$.
   Why: Power delivered equals voltage times current.

3. Energy increment is $dU = p\,dt = LI\,dI$.
   Why: Time cancels in the differential expression.

4. Integrate from zero to I: $U = L\int_0^I I'\,dI' = LI^2/2$.
   Why: L is constant and initial field energy is zero.

ANSWER $U = LI^2/2$ **for a linear inductor.**

Marks: 1 mark: law and assumptions; 4 marks: explanation or calculation; 1 mark: conclusion with units/limiting check.

Q31 Model · syllabus-based

Find the vacuum magnetic energy density for B=0.020 T, and the energy in 0.010 m³ of uniform field.

KNOW FIRST

T7 Magnetic field energy P1 School physics bridge M1 Mathematics from the beginning

1. $u_B = B^2/(2\mu_0)$.
   Why: The region is vacuum, so $H = B/\mu_0$.
2. $u_B = 0.0004/(8\pi \times 10^{-7}) = 159.2$ J/m³.
   Why: Square B before dividing.
3. $U = u_B V = 1.592$ J.
   Why: The energy density is uniform.

ANSWER $u_B = 159.2$ J/m³; U=1.592 J.

Marks: 1 mark: law and assumptions; 4 marks: explanation or calculation; 1 mark: conclusion with units/limiting check.

Q32 Model · syllabus-based

An ideal parallel-plate vacuum capacitor has area A and charging current I. Show that the displacement current through the gap equals I.

KNOW FIRST

T8 Charge conservation and Maxwell's repair P1 School physics bridge M1 Mathematics from the beginning

1. $E = Q/(\epsilon_0 A)$ between the plates away from edges.
   Why: Gauss's law gives the uniform field.

2. Electric flux through the plate area is $\Phi_E = EA = Q/\epsilon_0$.
   Why: Use the same area that the conduction-current loop spans.

3. $I_d = \epsilon_0 d\Phi_E/dt = dQ/dt = I$.
   Why: The change in charge creates the change in flux.

4. Thus a surface cutting the wire and a surface bulging through the gap give the same total Ampère-Maxwell contribution.
   Why: The extra term removes surface-choice inconsistency.

ANSWER $I_d = I$ **for the complete plate area.**

Marks: 1 mark: law and assumptions; 4 marks: explanation or calculation; 1 mark: conclusion with units/limiting check.

Q33 Legacy PYQ N25 Q6 · expanded to Maxwell system

Write the four integral Maxwell equations for a fixed loop/surface in vacuum, explaining the source or change in each.

KNOW FIRST

T9 The four Maxwell equations P1 School physics bridge M1 Mathematics from the beginning

1. $\oint_S E \cdot da = Q_{\text{enc}}/\epsilon_0$.
   Why: Enclosed electric charge produces electric flux.

2. $\oint_S B \cdot da = 0$.
   Why: No magnetic charge sources net closed-surface flux.

3. $\oint_C E \cdot dl = -d \int_S B \cdot da/dt$.
   Why: Changing magnetic flux produces electric circulation.

4. $\oint_C B \cdot dl = \mu_0 I_{\text{enc}} + \mu_0 \epsilon_0 d \int_S E \cdot da/dt$.
   Why: Both conduction and changing electric flux produce magnetic circulation.

5. In magnetostatics the displacement term vanishes, leaving $\oint B \cdot dl = \mu_0 I_{\text{enc}}$, the original integral Ampère law.
   Why: All fields are time-independent.

6. For curl laws, C is the boundary of S and their orientations obey the right-hand rule.
   Why: A consistent sign requires paired path and surface directions.

ANSWER **The two flux laws and two circulation laws form Maxwell's system.**

Marks: 1 mark: law and assumptions; 4 marks: explanation or calculation; 1 mark: conclusion with units/limiting check.

Q34 Model · syllabus-based

Show how changing polarization contributes to current and enters Ampère-Maxwell law in matter.

KNOW FIRST

T10 Maxwell equations in matter P1 School physics bridge M1 Mathematics from the beginning

1. A changing dipole density means charges shift locally, giving $J_p = \partial P/\partial t$.
   Why: Current includes bound charge motion.

2. Total current is $J = J_f + \nabla \times M + \partial P/\partial t$.
   Why: Magnetization loops and polarization changes are both bound contributions.

3. Insert this into vacuum-form Ampère-Maxwell and move curl M left.
   Why: Separate the material current terms.

4. The result is $\nabla \times H = J_f + \partial(\epsilon_0 E + P)/\partial t$.
   Why: Use the definition of H.

ANSWER $\nabla \times H = J_f + \partial D/\partial t$.

Marks: 1 mark: law and assumptions; 4 marks: explanation or calculation; 1 mark: conclusion with units/limiting check.

Q35 Model · syllabus-based

An interface has normal $\hat{n} = \hat{z}$ from medium 1 to 2 and free surface current $K_f = K\hat{x}$. What tangential H jump is required? Also state the normal B condition.

KNOW FIRST

T11 Electromagnetic boundary conditions P1 School physics bridge M1 Mathematics from the beginning

1. Write $\Delta H = H_2 - H_1$. The condition is $\hat{z} \times \Delta H = K\hat{x}$.
   Why: A narrow Amperian loop encloses the sheet current.

2. Since $\hat{z} \times \hat{y} = -\hat{x}$, $\Delta H_t = -K\hat{y}$.
   Why: Choose the component with the required cross product.

3. Normal B satisfies $B_{2z} = B_{1z}$.
   Why: A magnetic pillbox encloses no magnetic charge.

4. The normal H components require permeability information if the media differ.
   Why: B and H are not interchangeable.

ANSWER $H_{2t} - H_{1t} = -K\hat{y}$; **normal B is continuous.**

Marks: 1 mark: law and assumptions; 4 marks: explanation or calculation; 1 mark: conclusion with units/limiting check.

Q36 Model · syllabus-based

At a stationary interface with no free surface charge/current, medium 2 has $\epsilon_2 = 2\epsilon_1$, $\mu_2 = 3\mu_1$. Relate all normal and tangential E and H components.

KNOW FIRST

T11 Electromagnetic boundary conditions P1 School physics bridge M1 Mathematics from the beginning

1. $D_{2n} = D_{1n}$ gives $E_{2n} = E_{1n}/2$.
   Why: No free surface charge is enclosed by a thin pillbox.

2. $E_{2t} = E_{1t}$.
   Why: Faraday flux through a vanishing loop tends to zero for finite fields.

3. $B_{2n} = B_{1n}$ gives $H_{2n} = H_{1n}/3$.
   Why: Normal magnetic flux is continuous.

4. $H_{2t} = H_{1t}$.
   Why: There is no free sheet current.

ANSWER **Normal E and H scale inversely with ε and μ; tangential E and H are continuous here.**

Marks: 1 mark: law and assumptions; 4 marks: explanation or calculation; 1 mark: conclusion with units/limiting check.

Section C

Q37 Legacy PYQ N25 Q21 · expanded

Define line, surface and volume current measures, derive charge continuity, then explain Maxwell's correction to Ampère's law using a charging capacitor.

KNOW FIRST

T8 Charge conservation and Maxwell's repair P1 School physics bridge M1 Mathematics from the beginning

1. For line charge density λ moving along a wire at speed v, $I = \lambda v$ in A. Surface current density is $K = \sigma_q v$ in A/m, and volume current density is $J = \rho v$ in A/m² for a single moving charge population.
   Why: Each is charge density times velocity, with the appropriate cross-section dimension.

2. Charge leaving a fixed volume per time is $\oint J \cdot da = -d \int \rho \, d\tau / dt$.
   Why: Outward flow reduces the charge remaining inside.

3. The divergence theorem converts this to $\int (\nabla \cdot J + \partial_t \rho) d\tau = 0$. Since every volume satisfies it, $\nabla \cdot J = -\partial_t \rho$.
   Why: A continuous integrand whose integral vanishes in every volume must vanish locally.

4. The old law is $\nabla \times B = \mu_0 J$. Taking its divergence forces $\nabla \cdot J = 0$.
   Why: Divergence of curl vanishes identically.

5. Continuity demands $\nabla \cdot J = -\partial_t \rho$, which is nonzero where charge accumulates.
   Why: Local charge conservation must hold.

6. Gauss's law gives $\rho = \epsilon_0 \nabla \cdot E$, so $\partial_t \rho = \epsilon_0 \nabla \cdot \partial_t E$.
   Why: Differentiate the electric source law.

7. Adding $\epsilon_0 \partial_t E$ to J makes the total divergence zero.
   Why: The two divergence terms cancel.

8. The corrected law is $\nabla \times B = \mu_0 (J + \epsilon_0 \partial_t E)$.
   Why: It contains conduction and displacement-current density.

9. For a vacuum capacitor $E = Q/(\epsilon_0 A)$, giving $I_d = \epsilon_0 A \dot{E} = \dot{Q} = I$.
   Why: Both possible spanning surfaces now give identical magnetic circulation.

ANSWER **Maxwell's displacement current restores charge conservation and surface independence.**

Marks: 2 marks: definitions and assumptions; 5 marks: connected derivation/explanation; 2 marks: application or physical interpretation; 1 mark: final result and conditions.

Q38 Model · syllabus-based

Compare induction in a stationary loop with induction in a moving conductor, including the correct emf expression, sign and energy source.

KNOW FIRST

T5 Induced electric field P1 School physics bridge M1 Mathematics from the beginning

1. For a stationary loop, $\mathcal{E} = \oint E \cdot dl = -d\Phi_B/dt$.
   Why: Only the electric force contributes along a fixed path.

2. For a moving material loop, $\mathcal{E} = \oint (E + u \times B) \cdot dl$, where u is the conductor velocity.
   Why: Charges share the conductor's transport motion.

3. For an ordinary moving closed circuit, the combined expression equals the negative rate of linked flux.
   Why: Both field variation and changing surface geometry enter the flux derivative.

4. The loop orientation and area normal must be paired.
   Why: Faraday's sign is meaningful only with this convention.

5. Lenz's rule opposes the flux change. In a moving rod this creates drag, and the pulling work becomes electrical heating.
   Why: Energy conservation rules out self-amplifying motion.

6. In a stationary coil near a changing source field, the field-generating apparatus supplies the transferred energy.
   Why: A nonconservative E can transfer energy without mechanical motion of the receiving loop.

ANSWER **Stationary induction uses circulating E; motional induction includes $u \times B$; both obey the oriented flux rule under the stated conditions.**

Marks: 2 marks: definitions and assumptions; 5 marks: connected derivation/explanation; 2 marks: application or physical interpretation; 1 mark: final result and conditions.

Q39 Model · syllabus-based

Derive magnetic energy from circuit work and recover the energy density inside a long linear solenoid. Discuss assumptions.

KNOW FIRST

T7 Magnetic field energy P1 School physics bridge M1 Mathematics from the beginning

1. The incremental source work is $dU = LI\,dI$ for constant L.
   Why: The passive voltage is $L\dot{I}$.

2. Integration gives $U = LI^2/2$.
   Why: Start from zero current.

3. A long solenoid has $L = \mu N^2 A/\ell$ and $H = NI/\ell$.
   Why: Use its uniform interior field, neglecting ends.

4. Substitute to get $U = (\mu H^2/2)A\ell$.
   Why: Replace current and turns by H.

5. Divide by field volume $A\ell$ to get $u_B = \mu H^2/2 = B \cdot H/2 = B^2/(2\mu)$.
   Why: The medium is linear and isotropic.

6. In vacuum use μ_0. For nonlinear or hysteretic media, energy changes require the appropriate integral of $H \cdot dB$ and can include loss.
   Why: A constant proportionality no longer describes the material path.

ANSWER **In a linear lossless medium, $u_B = B \cdot H/2$; in vacuum, $u_B = B^2/(2\mu_0)$.**

Marks: 2 marks: definitions and assumptions; 5 marks: connected derivation/explanation; 2 marks: application or physical interpretation; 1 mark: final result and conditions.

Q40 Model · syllabus-based

Derive the four macroscopic electromagnetic boundary conditions at a stationary ordinary interface carrying free surface charge σf and free surface current Kf.

KNOW FIRST

T11 Electromagnetic boundary conditions P1 School physics bridge M1 Mathematics from the beginning

1. Take n from medium 1 to 2 and shrink a pillbox across the surface.
   Why: This fixes the cap orientations and jump signs.
2. Gauss's law for D leaves $(D_2 - D_1) \cdot n = \sigma_f$.
   Why: Side flux vanishes, but enclosed sheet charge stays proportional to cap area.
3. Magnetic Gauss law gives $(B_2 - B_1) \cdot n = 0$.
   Why: There is no magnetic surface charge.
4. A tiny rectangular loop in Faraday's law gives $n \times (E_2 - E_1) = 0$.
   Why: The magnetic-flux derivative tends to zero with loop area for bounded fields.
5. The H circulation law gives $n \times (H_2 - H_1) = K_f$.
   Why: The free sheet current survives while the displacement flux through the narrow loop vanishes.

6. Then use material laws, such as $D_i = \epsilon_i E_i$ and $B_i = \mu_i H_i$, to obtain jumps of other components.
   Why: Constitutive relations differ across the interface.

ANSWER **Normal D jumps by σf, normal B is continuous, tangential E is continuous, and tangential H jumps according to Kf.**

Marks: 2 marks: definitions and assumptions; 5 marks: connected derivation/explanation; 2 marks: application or physical interpretation; 1 mark: final result and conditions.